

University of Michigan

Heuristic Evaluation

Micaela Ciambone, Layilah Campbell, Seorin Jang, Dongsub Kim, Shenghao Mei

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Executive Summary

The study evaluates the usability of the USDA Agricultural Commodity Import Requirements portal through a structured heuristic evaluation, building on prior research that identified issues with search functionality, regularity clarity, and information organization. The ACIR portal serves a wide range of stakeholders, including importers, brokers, and government officials, and supports over 100,000 country community combinations. Given the complexity and restricted access to its user base, a systematic access interface level usability issues without requiring direct user recruitment.

The evaluation was conducted using six of Nielsen's ten usability heuristics as a guiding framework. Each team member independently reviewed the portal, focusing on core user tasks such as searching for and interpreting import requirements. Each issue identified was documented with its corresponding heuristics violation and assigned a severity rating on a 0-4 scale. Following independent evaluation, the team aggregated findings, removed duplicates, and established consensus severity ratings through a collaborative debrief. The process ensured both breadth and reliability in identifying usability problems while prioritizing those with the highest impact on user experience.

The evaluation uncovered over 60 usability issues, which were synthesized into five key findings based on severity and evaluator agreement. The most critical issue was the rigidity of the search function, which requires exact terminology and offers little flexibility significantly hindering task completion. Additionally, help resources were found to be fragmented and disconnected from user workflows, forcing users to navigate away from tasks to seek clarification. Other major findings included poor navigation and lack of system feedback, such as missing loading indicators as well as confusing regularity jargon and inconsistent labeling that can lead to misinterpretation of requirements.

Based on these findings, several actionable recommendations were proposed. First, improving search flexibility through more forgiving input handling and enhanced autocomplete functionality is crucial to reducing user errors. Second, centralizing help content within the workflow can improve accessibility and reduce cognitive load. Third, implementing clearer navigation cues, such as breadcrumbs and consistent link behaviors, would support user orientation. Finally, enhancing system feedback and simplifying regulatory language would improve clarity and trust in the system. Together, these recommendations provide a prioritized roadmap for improving the ACIR portal's usability, ultimately enabling users to more effectively and accurately navigate complex import regulations. These recommendations also align with the project's broader goal of translating prior research into feasible interface improvements that can be implemented within salesforce constraints while better supporting both novice and experienced users.

Introduction

Previous phases of this project, including interviews, a survey, and a comparative evaluation, identified recurring concerns around search rigidity, inconsistent regulatory language, and weak information hierarchy. While those methods captured user perspectives and benchmarked against competitors, none systematically assessed the portal interface against established usability principles. A heuristic evaluation addresses this gap by providing a structured, expert-driven inspection that identifies design-level problems without requiring user recruitment (Nielsen & Molich, 1990). The goal of this evaluation is to identify and prioritize usability violations in the

public-facing ACIR portal and produce actionable recommendations the ACIR team can address within the constraints of their Salesforce platform.

Methods

A heuristic evaluation is an inspection method in which evaluators independently review an interface against recognized usability principles and document violations (Nielsen & Molich, 1990). Research shows that individual evaluators typically catch only 20 to 51 percent of issues, but aggregating findings from three to five evaluators produces substantially better coverage.

We used all ten of Jakob Nielsen's usability heuristics (Nielsen, 1994) as our framework. Before beginning, the team held a calibration session where we reviewed several ACIR portal screens together and discussed what would and would not constitute a violation in this context. Each of the five team members then independently evaluated the portal, focusing on the core task of searching for and interpreting commodity import requirements. Evaluators documented each issue with its violated heuristic, location, and a severity rating on a 0-to-4 scale ranging from cosmetic to catastrophic. Team members did not view each other's findings until all individual evaluations were complete. We then aggregated findings in a group debriefing, merged duplicates, noted how many evaluators independently flagged each issue, and assigned consensus severity ratings to produce a prioritized list.

Findings/Recommendations

Across all five evaluations, we identified over 60 individual issues. After merging duplicates during debrief, we narrowed these to five key findings, each flagged by at least four of five evaluators. They are presented in priority order based on consensus severity and relevance to the project's research goals. Full individual and aggregated issue lists are in Appendix A.

Finding 1: Search requires exact terminology and offers no flexibility (Severity: 4, Evaluators: 5/5). The search interface requires users to enter both a country and commodity using the correct scientific name, common name, or synonym. If the input does not match the expected format, the search fails with minimal guidance. The autocomplete dropdown disappears when users type a full name, forcing them to backspace until it reappears. Users cannot search by country alone, save a search, or modify one variable without starting over.

Finding 2: Help content is scattered and disconnected from the task flow (Severity: 4, Evaluators: 5/5). Support resources are spread across the Help page, Videos and Guides Hub, homepage banner, glossary, and manual pages with no single entry point. Users must leave their current task to access any of these. Technical terms within CIR documents and search forms go unexplained unless the user navigates to a separate glossary.

Finding 3: Users lose their place and cannot easily go back (Severity: 3, Evaluators: 5/5). The portal lacks internal navigation cues. Some links open in a new tab while others stay in the same window, with no indicator distinguishing them. Using the browser back button after going several levels deep often causes users to lose their search state. There are no breadcrumbs showing the user's path.

Finding 4: The system provides little feedback during loading and transitions (Severity: 3, Evaluators: 4/5).

Pages frequently go blank for several seconds during transitions with no loading spinner, skeleton screen, or progress message. Users cannot tell whether the system is working or has failed.

Finding 5: Regulatory jargon and inconsistent labeling create confusion (Severity: 3, Evaluators: 4/5). Tile labels such as "Plants for Planting and Propagation" and "Plants and Plant Products Not for Propagation" use regulatory terminology unclear to importers and the general public. In one documented case, conflicting classification data between tiles led an officer to issue incorrect guidance.

Recommendation for Finding 1

The portal should implement a faceted search model that allows for single-variable adjustments and country-only queries. A persistent autocomplete system and "Save Search" feature would further reduce the high interaction cost of repetitive data entry.

Recommendation for Finding 2

Consolidate all manuals and tutorials into a single "Help Hub" to prevent users from hunting through fragmented resources. Integrating hover-over tooltips for technical terms directly within the workflow will provide essential guidance without interrupting the task.

Recommendation for Finding 3

Implement a persistent breadcrumb trail and dedicated internal navigation buttons to prevent session errors caused by the browser's back button. Standardizing link icons to distinguish between internal navigation and new tabs will also restore user control.

Recommendation for Finding 4

Replace blank transition screens with skeleton loaders or progress indicators to reassure users that the system is actively processing data. Additionally, add a clear confirmation message after feedback submission to verify the interaction was successful.

Recommendation for Finding 5

Add plain-language sub-labels to regulatory tiles to bridge the gap between technical jargon and general user understanding. Auditing the database for conflicting commodity classifications will ensure users receive consistent, accurate guidance regardless of the path taken.

Discussion

This evaluation confirms that the ACIR portal's usability issues are systematic barriers, not surface-level design flaws. By identifying over 60 issues and narrowing them to five prioritized findings, we moved beyond general user complaints toward a specific map of where the interface fails.

Study Shortcomings

Despite strong agreement among all five evaluators, this study has limitations common to expert-led inspections. The most significant is the absence of actual users. While we can confidently label a missing loading spinner as a visibility violation, we cannot capture the workarounds that experienced import specialists may have developed over years of use. Our severity ratings are informed estimates of impact, not measured data on error rates or task completion times.

Our evaluation was also restricted to the public-facing portal. Because ACIR operates within a protected Salesforce environment with different access levels, internal-facing hurdles for government officials could not be audited. We also did not perform a deep accessibility audit such as WCAG compliance. Given the government-mandated nature of this tool, accessibility remains a critical area this evaluation only touched upon.

Future Directions

To translate these findings into improvements, the next phase should move from problem identification to solution validation.

First, targeted usability testing should place high-severity issues like search rigidity and scattered help resources in front of real users. Observing importers navigating these pain points will provide the qualitative evidence needed to advocate for backend changes within the USDA.

Second, the project should prototype a more forgiving search interface, including improved autocomplete behavior and flexible query options. These designs should account for Salesforce platform constraints to ensure they are buildable. Although prototyping is outside the scope of this course, we agreed it would provide meaningful value to the client.

Third, a card-sorting exercise could help restructure the portal's information architecture by bringing the glossary and guides directly into the search workflow rather than keeping them in a separate hub.

Appendices

Appendix A: Citations

Nielsen, J., & Molich, R. (1990). *Heuristic evaluation of user interfaces*. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 249-256). ACM. <https://doi.org/10.1145/97243.97281>

Nielsen, J. (1994). *10 usability heuristics for user interface design*. Nielsen Norman Group. <https://www.nngroup.com/articles/ten-usability-heuristics/>